

COMPUTER VISION · CO1 · TECHNICAL PRESENTATION

Image Formation Models

Pinhole Camera, Perspective Projection, Distortion & Illumination

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From a 3D World to a 2D Image

Every Computer Vision system begins with a physical problem: light bouncing off a 3D scene must be geometrically and optically mapped onto a flat 2D sensor — with real hardware that isn't perfect.

1

Geometric Mapping

How do 3D coordinates (X, Y, Z) become 2D pixel coordinates (u, v)?

2

Lens Distortion

Real lenses bend straight lines — barrel and pincushion effects.

3

Illumination Variation

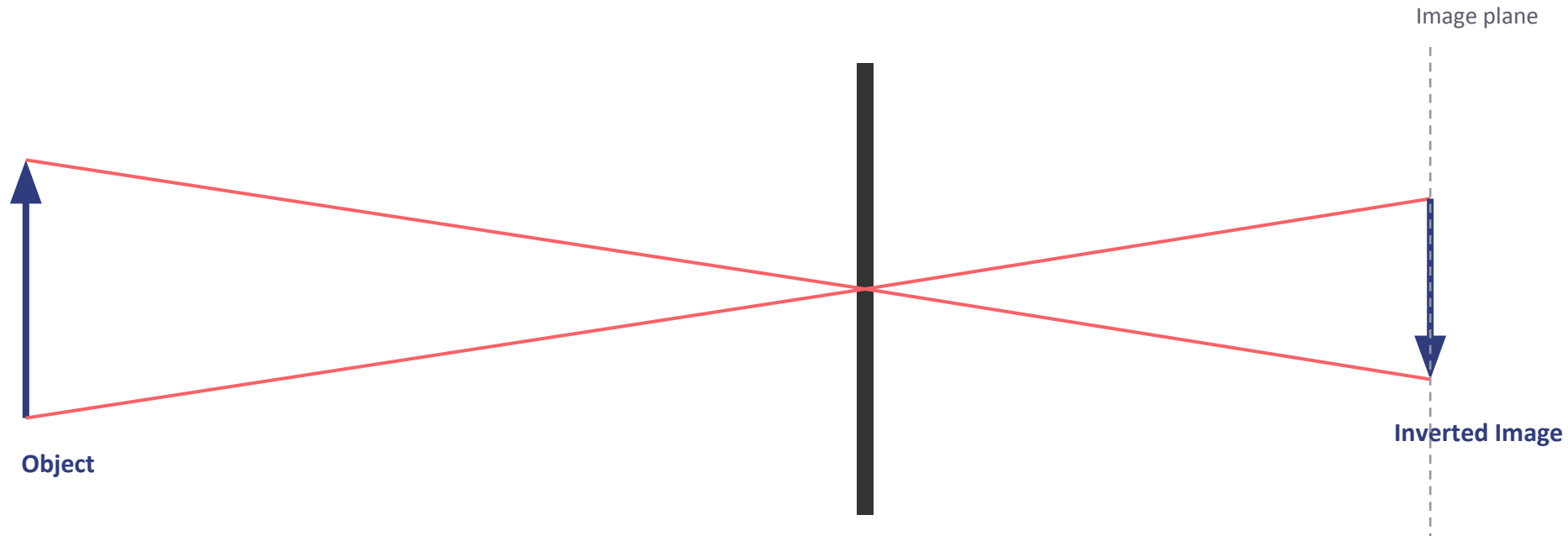
The same object looks different under different lighting conditions.

4

Depth Ambiguity

A single 2D image loses the depth information of the original scene.

The Pinhole Camera Model



How it works: Light from every point on the object travels in a straight line through a single small aperture (the pinhole), projecting an inverted, reversed image onto the plane behind it. This is the simplest possible camera — no lens, no focusing — and it is the theoretical basis for the projection equations used throughout Computer Vision.

Perspective Projection Geometry

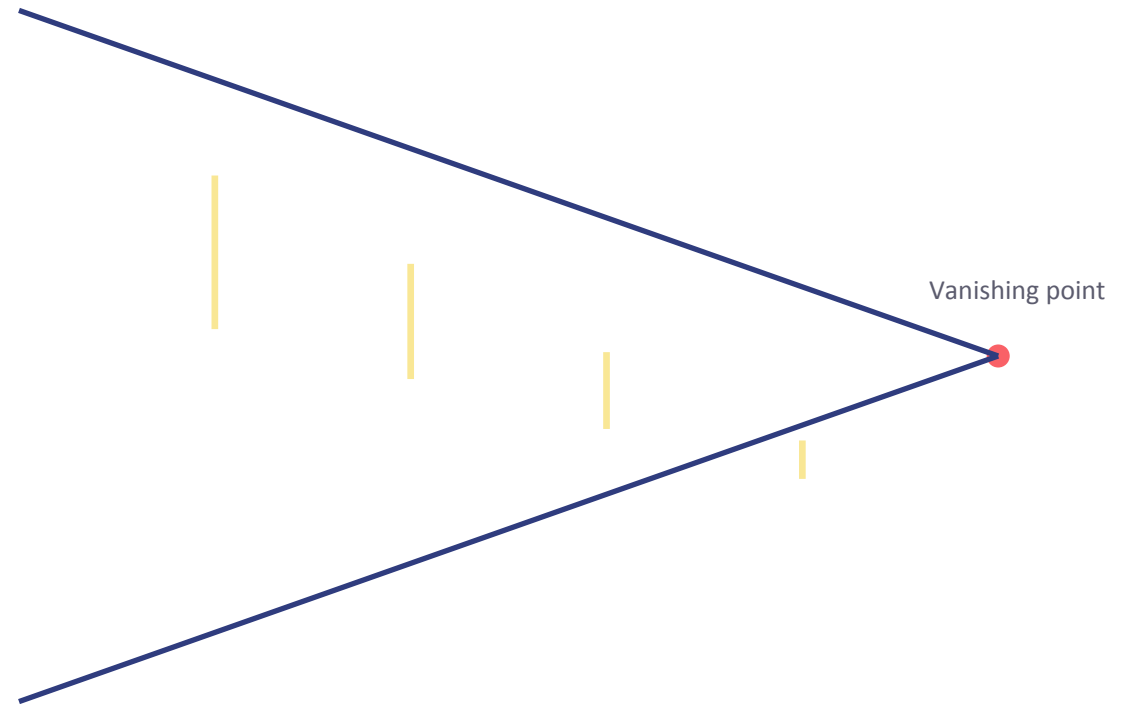
The Governing Equations

A 3D scene point (X, Y, Z) projects onto the 2D image plane at focal length f as:

$$x = f \cdot X / Z \quad y = f \cdot Y / Z$$

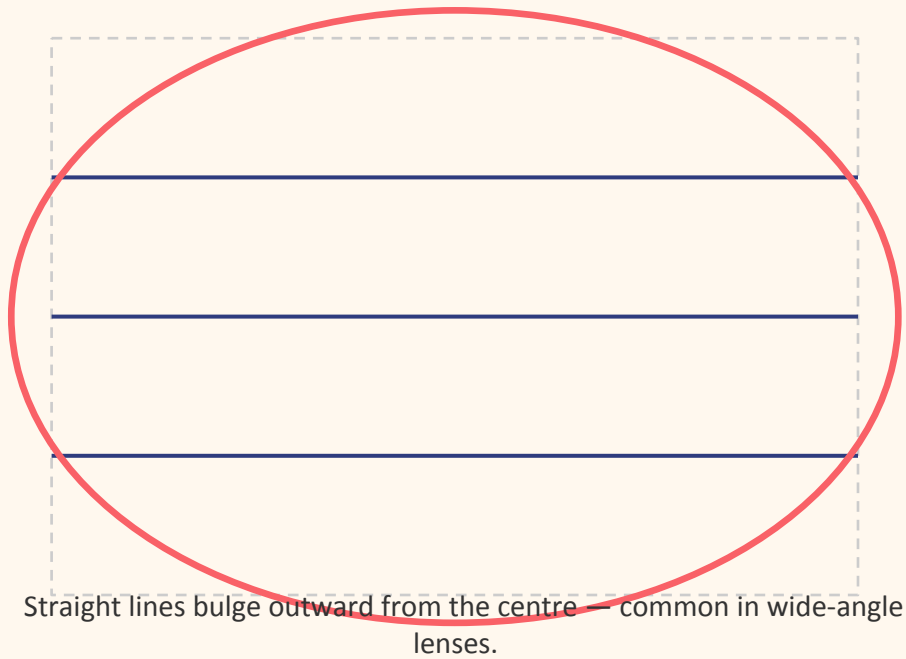
The division by depth Z is the mathematical signature of perspective: objects farther away (larger Z) project to smaller image coordinates.

Consequence: parallel lines in the world (e.g. railway tracks) appear to converge toward a vanishing point in the image.

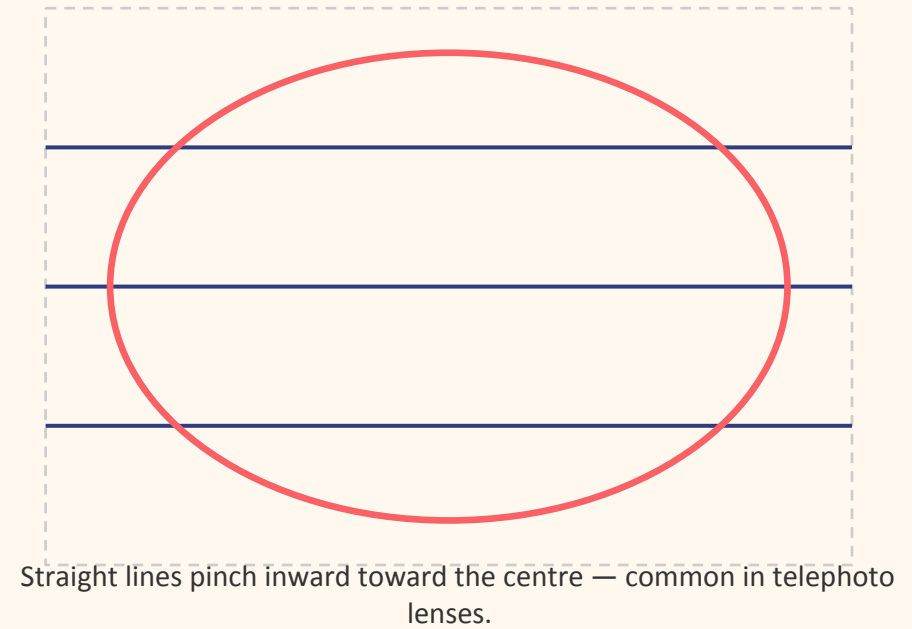


Lens Distortion

Barrel Distortion



Pincushion Distortion



Illumination Effects on Image Formation

Shading & Shadows

The same 3D geometry produces very different pixel intensities under different light directions.

Specular Highlights

Shiny surfaces create bright reflective spots that are view-dependent, not material-dependent.

Colour Temperature Shift

Indoor tungsten vs outdoor daylight changes the apparent colour of the same object.

Dynamic Range Limits

Sensors struggle to capture both bright highlights and dark shadows in the same frame.

Why This Matters for CV Systems

1

Camera Calibration

Estimating focal length and lens-distortion coefficients so 2D pixels can be mapped back to accurate 3D geometry.

2

Robust Feature Matching

Algorithms must tolerate illumination changes to reliably match the same point across images.

3

3D Reconstruction / SLAM

Perspective projection equations are inverted (triangulation) to recover depth from multiple views.

4

Autonomous Systems

Self-driving cars and robots must correct distortion and illumination effects in real time to perceive geometry correctly.

Conclusion

- 1 The pinhole camera model provides the simplest geometric explanation of how a 3D scene becomes a 2D image.
- 2 Perspective projection ($x = fX/Z$, $y = fY/Z$) mathematically captures why distant objects appear smaller.
- 3 Real lenses deviate from this ideal through barrel/pincushion distortion, which calibration must correct.
- 4 Illumination is not incidental — it fundamentally changes pixel values for identical geometry, and CV systems must be built to tolerate it.